

C02	Apply	PO1, PO2, PO3, PO4, PO5, PO7, PO9	L3	Examination, Assignment
C03	Analyze	PO1, PO2, PO3, PO4, PO5, PO7, PO9, PO10	L4	Assignments, Discussion, Examination
C04	Create	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12	L6	Assignments, Discussion, Examination
C05	Evaluate	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12	L5	Assignments, Discussion, Examination

CO-PO/PSO Mapping: (3-Strong Correlation 2- Medium Correlation 1- Low Correlation)

Course Outcomes	Blooms Taxonomy Level	Programme Outcomes (PO)												Programme Specific Outcomes (PSO)			
		PO 01	PO 02	PO 03	PO 04	PO 05	PO 06	PO 07	PO 08	PO 09	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
C01	L2	3	2	-	-	-	-	3	-	2	-	-	-	3	1	1	1
C02	L3	3	2	2	3	3	-	3	-	2	-	-	-	3	2	2	2
C03	L4	3	3	3	3	3	-	3	-	2	2	-	-	3	3	2	2
C04	L6	3	3	3	3	3	3	3	2	2	3	2	2	3	3	3	3
C05	L5	3	3	3	3	3	3	3	2	2	2	2	2	3	3	3	3
		3	2.6	2.8	3	3	3	3	2	2	2.3	2	2	3	2.4	2.2	2.2

Course Contents:

Module	Details	Contact Hours
I	Introductory Concepts Overview of Data Science and Data Analytics, Types of Analytics: Descriptive, Diagnostics, Predictive and Prescriptive; Data Ubiquity, Nature of Data: Structured, Unstructured, Big Data; Advantages of Data-Driven Decisions, Data Science Process, Applications of Data Science in various fields, Data Science Roles, Data Security, Privacy, and Ethical Issues.	09
II	Data Preparation Data Collection Methods: Primary, Secondary data; Pre-Processing: Data Cleaning, Data Integration and Transformation, Data Discretization; Dimensionality Reduction, PCA, Feature Engineering and Selection.	09
III	Exploratory Data Analysis What is Exploratory Data Analysis (EDA)? Descriptive Statistics: Mean, Standard Deviation, Skewness and Kurtosis; Types of EDA: Univariate, Bi-Variate, Multi-Variate; Visualizing EDA: Histogram, Scattered Plot, Box Plots,	09

	Pivot Table, Heat Map; Correlation Statistics, Statistical Significance and Hypothesis Testing, ANOVA.	
IV	Machine Learning and Model Building Machine Learning in Data Science, Types of Machine Learning, Supervised, Unsupervised, Modelling Process, Simple and Multiple Regression, Time Series Analysis, Classification, Prediction, Clustering, Decision Tree, k-Nearest Neighbour, Association Rules Mining.	09
V	Model Evaluation Model Accuracy, Overfitting, Under Fitting, Bias-Variance Trade-off, Evaluation Metrics, Confusion Matrix, Methods for evaluating Accuracy, Cross Validation, Model Evaluation using Visualization: Residual Plot, Distribution Plot, ROC curve, Model Selection.	09

Text Book (TB):

1. Data Science: Concepts and Practice, by Vijay Kotu and Bala Deshpande, Morgan Kaufmann Publications, 2019.
2. A General Introduction to Data Analytics, by João Moreira, Andre Carvalho, Tomás Horvath, John Wiley & Sons, 2019.

Reference Books (RB):

1. Introducing Data Science, by Davy Cielen, Arno D. B. Meysman, Mohamed Ali, Manning Publications, 2016
2. Data Science for Business, by Foster Provost and Tom Fawcett, O'Reilly, 2013.
3. Data Science and Big Data Analytics by EMC Education Services (Editor), Wiley, 2015
4. Doing Data Science by Rachel Schutt and Cathy O'Neil, O'Reilly, 2014
5. Practical Statistics for Data Scientists by Peter Bruce and Andrew Bruce, O'Reilly, 2017.

Web Video Links:

1. <https://www.linkedin.com/learning/data-science-foundations-fundamentals-5/the-fundamentals-of-data-science?autoAdvance=true&autoSkip=false&autoplay=true&resume=false&u=92695330>
2. <https://www.linkedin.com/learning/introduction-to-data-science-2/beginning-your-data-science-exploration?autoAdvance=true&autoSkip=false&autoplay=true&resume=true&u=92695330>
3. <https://www.linkedin.com/learning/paths/master-excel-for-data-science>

Web Text Links:

1. <https://towardsdatascience.com/the-evolution-of-data-science-as-i-remember-it-54cf4e602bd5>

Abbreviations & Expansions			
Pedagogy/Activity planned		Mode of Delivery	
P1	PPT presentation & Classroom interaction	M1	Synchronous – PPT
P2	Blended Learning with Hands on	M2	Asynchronous / synchronous
P3	Flip Class & Module Quiz	M3	Synchronous - Hands on
A1	Activity-1	M4	Synchronous - Discussion
A2	Activity-2		